

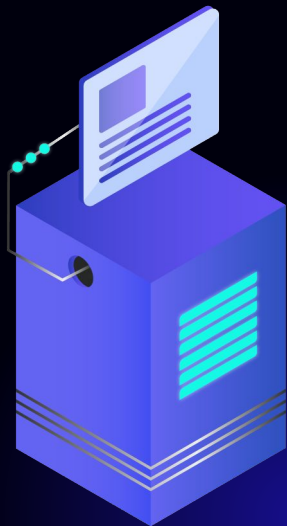
Cloud Workload Execution Models

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INTRODUCTION

- A Cloud Computing System with finite capacity **C** serving **N** Users
- Each User generates a new Workload every **T** seconds
- Every Workload requires a Computing Capacity **R** for **D** seconds
- If Capacity is not enough $\rightarrow$ Workload enters a **FCFS Queue**

Two Models

Container Based

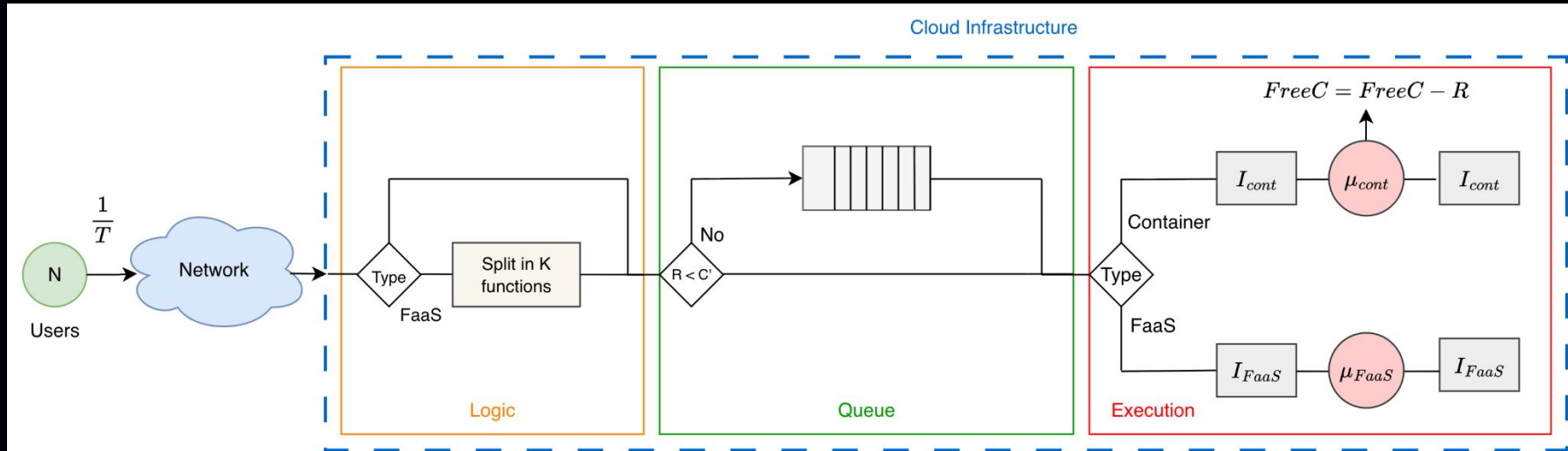
- Monolithic
- Only set up once
- Higher initialization and destruction cost $\rightarrow I_{\text{Cont}}$

Function as a Service

- Workload gets split into **K** Sequential Functions
- Lower initialization and destruction cost per function $\rightarrow I_{\text{FaaS}} < I_{\text{Cont}}$
- Cumulative Overhead

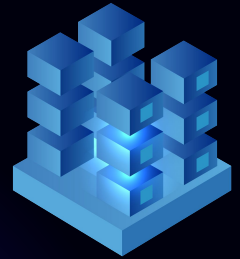
OBJECTIVES

- Evaluating the total execution time of workloads for various values of N and K.
- Comparing the performance between container-based model and Function as a Service model.



Scheme of the System

SYSTEM MODELING

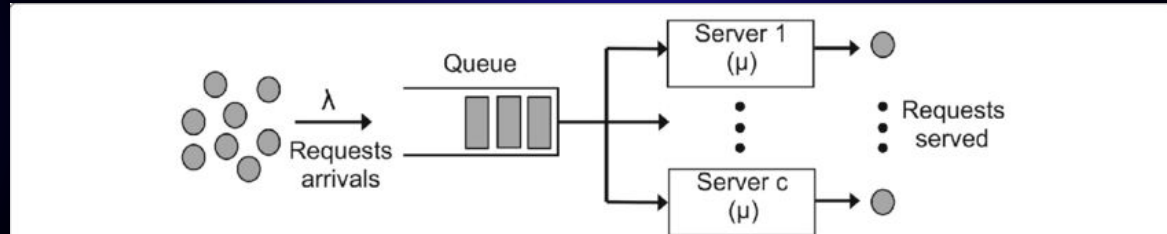


Assumptions

- Capacity C is fixed and shared among all the Users
- No Workload loss due to communication channel
- Overheads I_{Cont} and I_{FaaS} do not consume any additional capacity to R
- The type of Workload generated by a User is determined using a uniform random distribution

Theoretical Model

- We chose a M/M/c model for our system
- Real model approximates M/G/c due to overheads
- Real-life capacity C is shared among servers

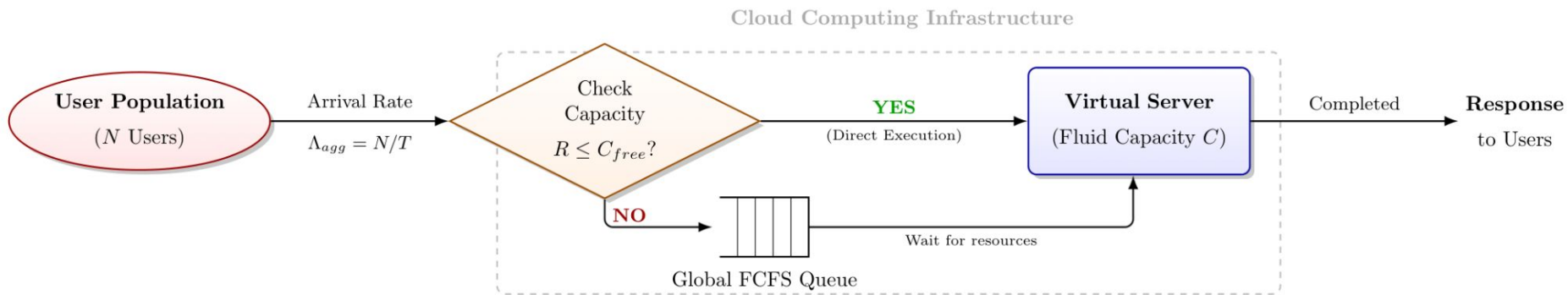
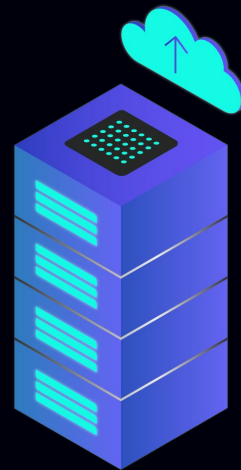


Representation of a M/M/c system

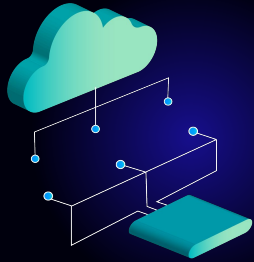
IMPLEMENTATION

Simple but Real

- Single Aggregate Workload Source to implement the User Population
- Virtually Infinite Buffer
- Single Virtual Server typical of Cloud Computing scenarios
- Aggregate Capacity C is fluid and continuous



Implemented System Scheme



VERIFICATION

01

Degeneracy Test

Evaluating the Simulator's robustness under limit conditions

- Idle State and Extreme Saturation ($\lambda \gg \mu$)
- No Capacity and Unlimited Capacity
- Infinite Arrival Rate
- $R > C$
- Only Container Mode
- Only FaaS Mode
- Results are consistent

Validation through Theoretical model

Checking the correctness of the Simulator's queueing logic

- Used analytical equations for M/M/c systems
- Set overheads to zero ($I_{\text{Cont}} = 0 ; I_{\text{FaaS}} = 0$)
- Compared Mean Response Time
- $R > C$
- Theoretical value falls into the confidence interval derived from simulations

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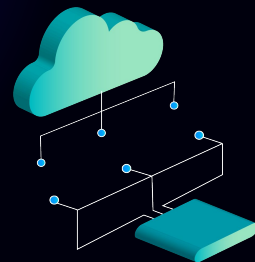
Metric	Value
Theoretical Mean Response Time ($E[R]$)	5.07574 s
Simulated Mean ($\bar{X}$)	5.07991 s
Standard Deviation (S)	0.03387 s
Sample Size (<i>runs</i>)	50
95% Confidence Interval	[5.07052, 5.08930] s

Theoretical Validation: numerical results

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Consistency Test

Evaluating the Simulator's coherence when scaling parameters

- Config. A $\rightarrow E[T] = 5s ; E[D] = 2.5s$
- Config. B $\rightarrow E[T] = 10s ; E[D] = 5s$
- Disabled overheads for pure theoretical evaluation ($I_{Cont} = 0 ; I_{FaaS} = 0$)
- Utilization and Avg. Queue Length shouldn't change
- Waiting Time and Response Time should double up

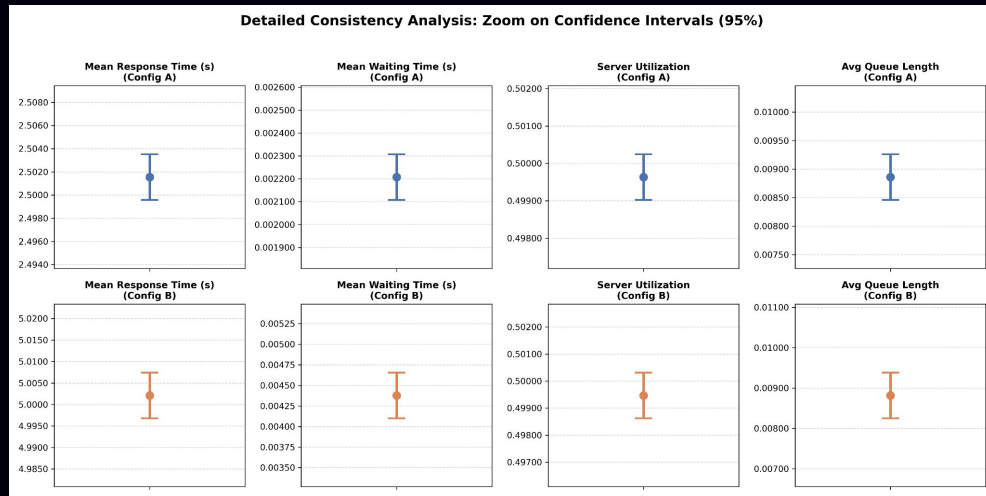
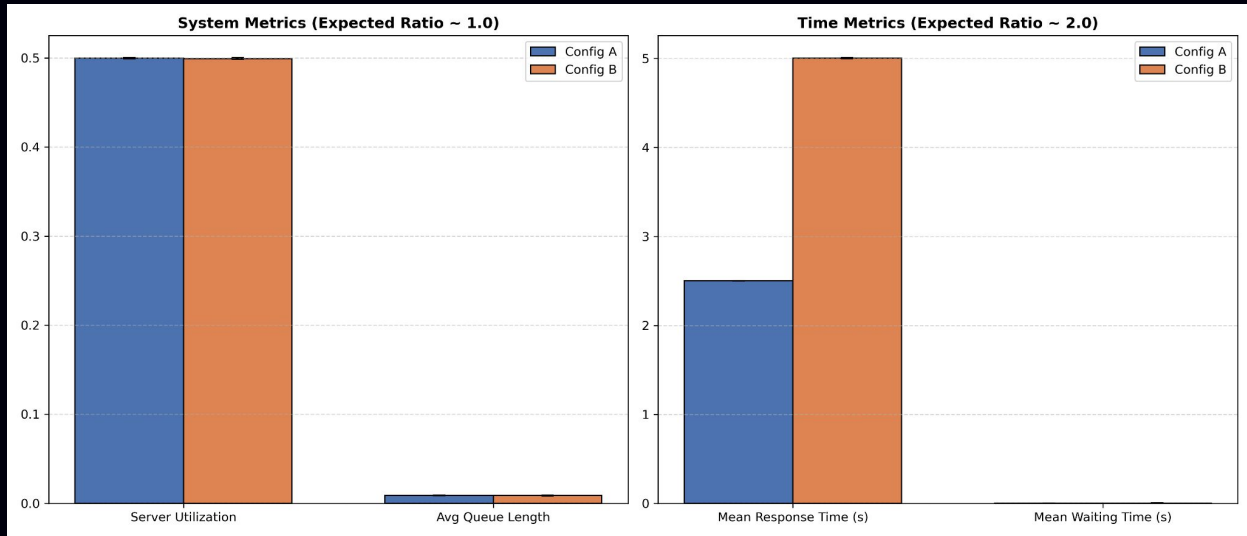
Continuity Test

Checking that small input changes result in small predictable output variations

- Medium-High System Load (72%)
- Config. A $\rightarrow N = 64$ Users
- Config. B $\rightarrow N = 65$ Users
- Adopted **Paired Observations** analysis method
- Results proved Continuity and Monotonicity

VERIFICATION

Consistency Test

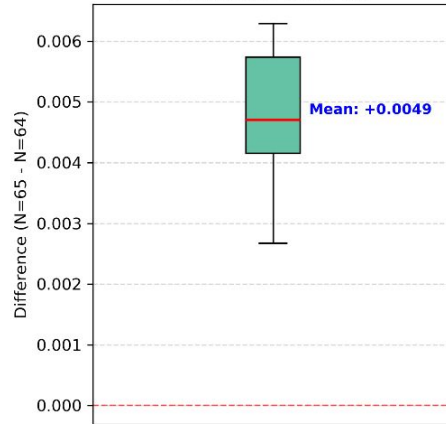


VERIFICATION

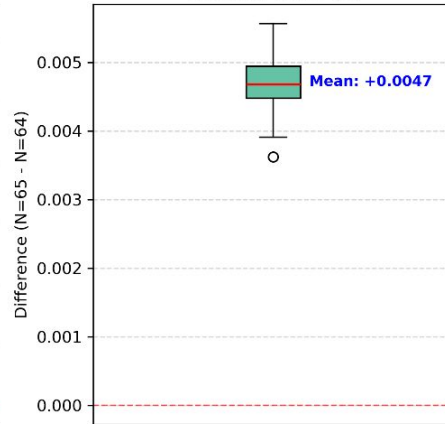
Continuity Test

Continuity Test: Paired Differences Distribution (N=65 vs N=64)
Positive values indicate monotonicity (Increase with load)

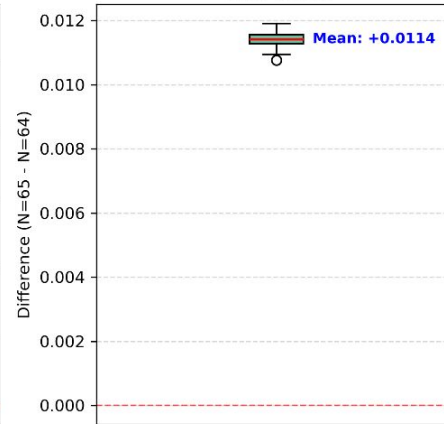
Response Time (s)



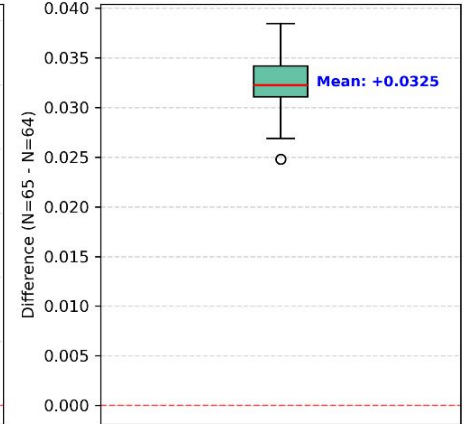
Waiting Time (s)



Utilization



Queue Length



CALIBRATION

- The Dimensioning of our System was performed by analyzing various scientific papers as references to real-world systems
- Parameters were scaled for practical simulation necessities (scaling factor $\alpha = 1000$)

<i>Parameter</i>	<i>Reference Scenario</i>	<i>Scaled Simulation</i>
<i>Total Capacity (C)</i>	40,000 GHz	40 GHz
<i>Total vCPUs (k)</i>	20,000	20
<i>System Users (N)</i>	[5k ; 100k]	20 ; 90

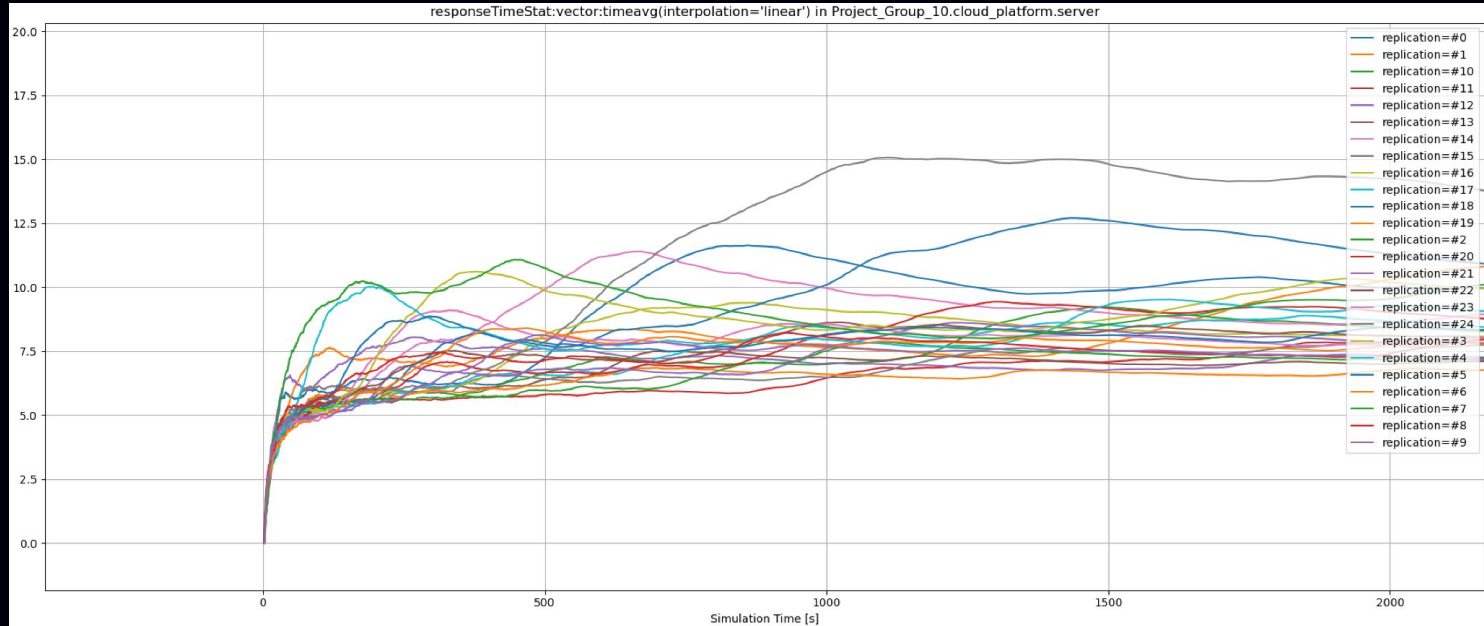
<i>Parameter</i>	<i>Value</i>
<i>I_{Cont}</i>	0.800 s
<i>I_{FaaS}</i>	0.125 s

Overheads

CALIBRATION

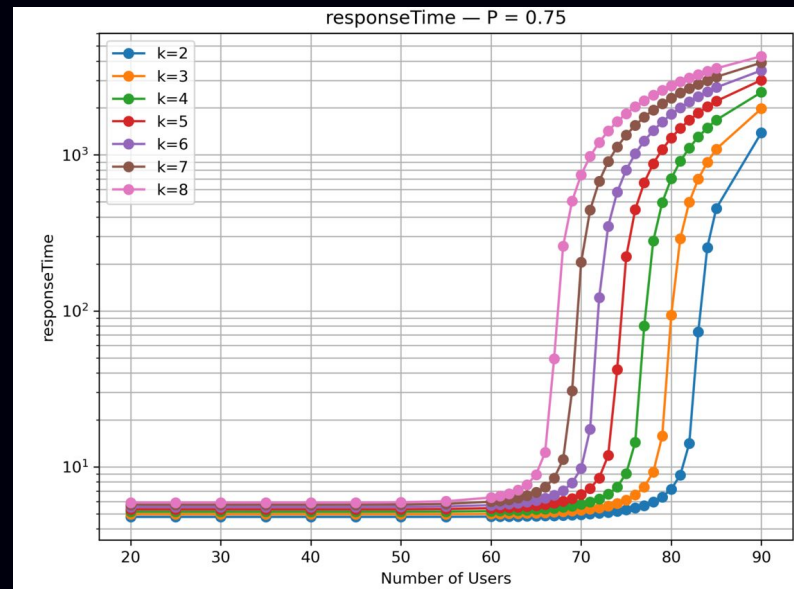
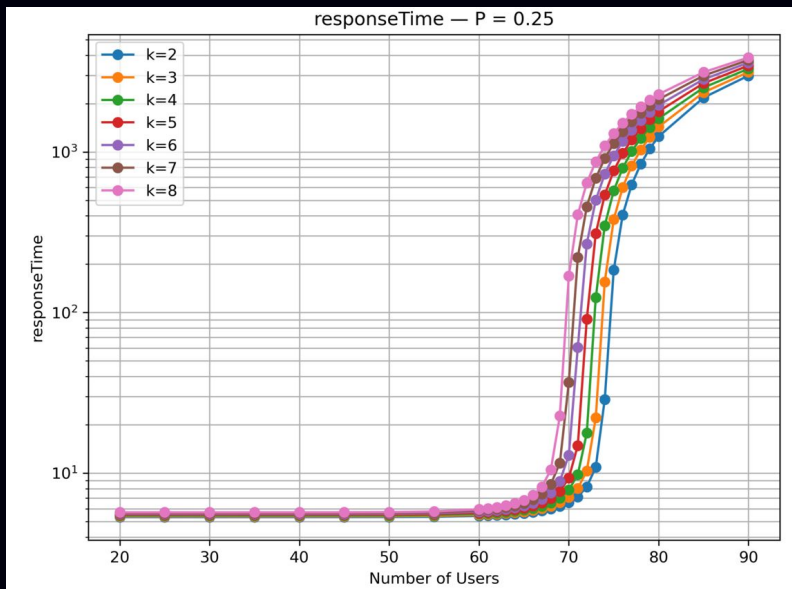
Warm-up Time

- Calibrated using **Response Time**
- Set to 500 s



Response Time

- Used different balance between Container-mode and FaaS-mode workloads in the system ($p = 0.25$; $p = 0.75$)
- Showed the System's behaviour as the number of users increases
- Displayed the impact of different FaaS granularity (K)



System's Utilization

- Used different balance between Container-mode and FaaS-mode workloads in the system ($p = 0.25$; $p = 0.75$)
- Utilization increases linearly, but FaaS workload proportion changes steepness
- System approaches saturation faster when granularity is higher (K)

